

```

import java.util.ArrayDeque;
import java.util.Scanner;
public class Main
{
    public static void main(String[] args)
    {
        ArrayDeque<String>patients = new ArrayDeque<>();

        Scanner sc = new Scanner(System.in);
        int choice;
        do{
            System.out.println("HOSPITAL MANAGEMENT SYSTEMM");
            System.out.println("1. add emergency");
            System.out.println("2. add normal");
            System.out.println("3.treat emergency first");
            System.out.println("4.remove last patient");
            System.out.println("5.view firstpatient");
            System.out.println("6.View Last Patient");
            System.out.println("7. Display Queue");
            System.out.println("8. Queue Size");
            System.out.println("9. Exit");
            System.out.print("Enter your choice: ");
            choice = sc.nextInt();
            sc.nextLine();
            switch(choice)
            {
                case 1:
                    System.out.println("add emergency patient first");
                    String emergency = sc.nextLine();
                    patients.addFirst(emergency);
                    System.out.println("Emergency patient added successfully.");
                    break;
                case 2:
                    System.out.println("add normal patient");
                    String normal = sc.nextLine();
                    patients.addLast(normal);
                    System.out.println("normal patient added successfully.");
                    break;
                case 3:
                    if (patients.isEmpty()) {
                        System.out.println("No patients in the queue.");
                    } else {
                        System.out.println("Patient Treated: " + patients.removeFirst());
                    }
                    break;
                case 4:
                    if (patients.isEmpty()) {
                        System.out.println("No patients in the queue.");
                    } else {
                        System.out.println("Removed Last Patient: " + patients.removeLast());
                    }
                    break;
                case 5:
                    if (patients.isEmpty()) {
                        System.out.println("Queue is empty.");
                    } else {
                        System.out.println("First Patient: " + patients.peekFirst());
                    }
                    break;
                case 6:
                    if (patients.isEmpty()) {
                        System.out.println("Queue is empty.");
                    } else {
                        System.out.println("Last Patient: " + patients.peekLast());
                    }
                    break;
                case 7:
                    if (patients.isEmpty()) {
                        System.out.println("Queue is empty.");
                    } else {
                        System.out.println("\nPatients in Queue:");
                        for (String patient : patients) {
                            System.out.println(patient);
                        }
                    }
            }
        }
    }
}

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        break;

    case 8:
        System.out.println("Total Patients: " + patients.size());
        break;
    case 9:
        System.out.println("Thank you");
        break;

    default:
        System.out.println("Invalid choice.");
    }

}

while (choice != 9);

sc.close();
}
```